

# DEGREE-SEVEN RIGIDITY UNDER $K_7^-$ -MINOR EXCLUSION

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**ABSTRACT.** Let  $G$  be a minor-minimal non-six-colourable graph with no  $K_7^-$  minor. We prove that  $G$  contains at most one  $K_5$  subgraph and that every degree-seven vertex  $v$  satisfies  $G[N(v)] \cong K_4 \dot{\cup} K_3$ . A private-triangle Kempe-chain argument further shows that no  $K_5$  contains five degree-seven vertices. Consequently, if  $n_i$  counts the vertices of degree  $i$ ,  $b$  counts the degree-eight vertices contained in no  $K_5$ , and  $\tau = \sum_{i \geq 10} (i - 9)n_i$ , then  $n_7 \leq 4$ ,  $|E(G)| \geq 4|V(G)| - 2$ , and  $b \geq 17 + \tau$ . More precisely, if  $n_7 \leq 3$ , then  $|E(G)| \geq 4|V(G)| - 1$ , while if  $n_7 = 4$ , then  $|V(G)| \geq 37$  and  $b \geq 32 + \tau$ . As a preliminary tool, we give a direct uniform proof that, for every  $r \geq 2$ , an  $(r + 1)$ -connected graph containing two distinct  $K_r$  subgraphs has a  $K_{r+2}^-$  minor.

## 1. INTRODUCTION

Hadwiger's conjecture states that every graph of chromatic number  $t$  contains a  $K_t$  minor [3]. The conjecture is open for  $t = 7$ . Write  $K_t^-$  for the graph obtained from  $K_t$  by deleting one edge, and write  $K_7^\vee$  for the graph obtained from  $K_7$  by deleting two incident edges. Every  $K_7^-$ -minor-free graph is seven-colourable [1, Theorem 1]; Albar's proof uses Jakobsen's extremal theorem. Norin and Totschnig proved that every  $K_7^\vee$ -minor-free graph is six-colourable [8, Theorem 4]. They asked whether the same conclusion holds under the weaker exclusion of  $K_7^-$  [8, Conjecture 21].

Neighbourhoods of low-degree vertices in contraction-critical graphs have a long history. Mayer obtained a finite catalogue of candidate neighbourhood graphs for degree-seven vertices in six-contraction-critical graphs [7]. In the minor-minimal  $K_7^\vee$ -minor-free setting, Norin and Totschnig proved that every degree-seven vertex belongs to a  $K_5$  contained in its closed neighbourhood [8, Claim 4.4]. Here a rooted-minor argument determines the entire neighbourhood and, after the linked-clique reduction below, leaves only  $K_4 \dot{\cup} K_3$ .

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With substantial mathematical contributions from GPT-5.6 Sol, an OpenAI model.

We first state the unconditional linkage theorem used in that reduction. Rolek and Song proved its case  $r = 6$  [9, Lemma 1.9]; the argument below gives a direct uniform proof.

**Theorem 1.1.** *For every  $r \geq 2$ , every  $(r + 1)$ -connected graph containing two distinct  $K_r$  subgraphs contains a  $K_{r+2}^-$  minor.*

For  $2 \leq r \leq 6$ , Theorem 1.1 also follows from Rolek and Song's result. Adjoin a universal clique of order  $6 - r$ . The resulting graph is seven-connected and contains two distinct  $K_6$  subgraphs: deleting at most six vertices either leaves an added universal vertex or removes the whole added clique and at most  $r$  vertices of the original graph. Their lemma therefore gives a  $K_8^-$  model. Since the model bags are disjoint, at most  $6 - r$  of them meet the added clique. Discard those bags and retain any  $r + 2$  of the remaining bags; they form a  $K_{r+2}^-$  model. We include the direct proof because it applies for every  $r \geq 2$  and gives the localisation in Lemma 2.1. The connectivity hypothesis is sharp:  $K_{r+1}$  is  $r$ -connected and contains two distinct  $K_r$  subgraphs, but has no  $K_{r+2}^-$  minor.

Our second result describes the low-degree structure of a hypothetical counterexample to the Norin–Totschnig question. A graph is *minor-minimal non-six-colourable* if it is not six-colourable and every proper minor is six-colourable.

**Theorem 1.2.** *Let  $G$  be a minor-minimal non-six-colourable  $K_7^-$ -minor-free graph. For each  $i$ , let  $n_i$  denote the number of vertices of degree  $i$ . Let  $b$  denote the number of degree-eight vertices contained in no  $K_5$  subgraph; equivalently,  $b$  counts the degree-eight vertices  $v$  for which  $G[N(v)]$  contains no  $K_4$ . Put*

$$\tau = \sum_{i \geq 10} (i - 9)n_i.$$

*Then  $G$  is seven-connected, contains at most one  $K_5$  subgraph, and*

$$G[N(v)] \cong K_4 \dot{\cup} K_3 \quad \text{for every vertex } v \text{ of degree seven.}$$

*No  $K_5$  subgraph contains five degree-seven vertices. Moreover, one of the following holds:*

- (1)  $n_7 \leq 3$ ,  $|E(G)| \geq 4|V(G)| - 1$ , and  $b \geq 17 + \tau$ ;
- (2)  $n_7 = 4$ ,  $|V(G)| \geq 37$ , and  $b \geq 32 + \tau$ .

*In particular,*

$$n_7 \leq 4, \quad |E(G)| \geq 4|V(G)| - 2, \quad b \geq 17 + \tau.$$

All proofs below are computation-free.

A *minor model* of a graph  $H$  in  $G$  is a family  $(B_x : x \in V(H))$  of pairwise disjoint connected vertex sets such that  $B_x$  and  $B_y$  are adjacent whenever  $xy \in E(H)$ . We use  $H \preceq G$  to denote that  $H$  is a minor of  $G$ . All graphs are finite and simple.

## 2. TWO LINKED CLIQUES

We begin with the disjoint case.

**Lemma 2.1.** *Let  $r \geq 2$ . If an  $(r+1)$ -connected graph  $H$  contains two vertex-disjoint  $K_r$  subgraphs, then  $K_{r+2}^- \preceq H$ . The branch sets can be chosen to meet the union of the two cliques.*

*Proof.* Write the two cliques as

$$A = \{a_1, \dots, a_r\}, \quad B = \{b_1, \dots, b_r\}.$$

By Menger's theorem there are  $r$  pairwise vertex-disjoint  $A$  to  $B$  paths whose open interiors avoid  $A \cup B$ . Relabel them as paths  $P_i$  from  $a_i$  to  $b_i$ , and put  $U = \bigcup_i V(P_i)$ .

Call a path  $Q$  a *usable cross-path* if its ends lie on distinct paths  $P_i, P_j$ , its open interior avoids  $U$ , and its ends are not both  $A$ -ends or both  $B$ -ends. Suppose first that  $Q$  exists. Split each of  $P_i$  and  $P_j$  across an edge into a nonempty  $A$ -prefix and a nonempty  $B$ -suffix, choosing the split so that the ends of  $Q$  belong to pieces of opposite kinds. Absorb the open interior of  $Q$  into the piece containing one end. The four resulting pieces are connected and disjoint. The two  $A$ -pieces are adjacent through  $A$ , the two  $B$ -pieces are adjacent through  $B$ , the two pieces from each original path are adjacent across its splitting edge, and  $Q$  supplies one of the two remaining cross-adjacencies. Thus at most one adjacency among the four pieces is missing. Each unsplit path is adjacent to every split piece through its two clique ends, and the unsplit paths are pairwise adjacent. These  $r+2$  sets form a  $K_{r+2}^-$  model.

Suppose now that no usable cross-path exists. Let  $C$  be a component of  $H - U$ . If  $C$  has neighbours on two distinct paths, a path through  $C$  between two such attachments is unusable only when both attachments are  $A$ -ends or both are  $B$ -ends. Comparing all pairs of attachments shows that all neighbours of  $C$  are  $A$ -ends or all are  $B$ -ends. Hence  $|N_H(C)| \leq r$ , contrary to  $(r+1)$ -connectivity. Every component of  $H - U$  therefore attaches to only one  $P_i$ .

For each  $i$ , let  $W_i$  be the union of the open interior of  $P_i$  and all components of  $H - U$  that attach to  $P_i$ . No vertex of  $W_i$  has a neighbour on  $P_j$  for  $j \neq i$ . Indeed, an edge from the open interior of  $P_i$  to  $P_j$  would be a usable cross-path, while such an edge from a component of  $H - U$  would make that component attach to two distinct linkage paths, contrary to the preceding paragraph. Hence

$$N_H(W_i) \subseteq \{a_i, b_i\}.$$

Since  $r \geq 2$ , a nonempty  $W_i$  would give a vertex cut of order at most two. Thus every  $W_i$  is empty. It follows that  $H$  consists of  $A$ ,  $B$ , and the matching edges  $a_i b_i$ : any other edge between  $A$  and  $B$  would be usable. This graph has minimum degree  $r$ , again contradicting  $(r+1)$ -connectivity.  $\square$

*Proof of Theorem 1.1.* Let  $L_1, L_2$  be distinct  $K_r$  subgraphs and put  $s = |V(L_1) \cap V(L_2)|$ . If  $s \leq r - 2$ , delete the common set  $Z$ . The graph  $G - Z$  is  $(r + 1 - s)$ -connected and contains the two disjoint  $K_{r-s}$  subgraphs  $L_1 - Z$  and  $L_2 - Z$ . Lemma 2.1 gives a  $K_{r-s+2}^-$  model in which every branch set contains a vertex of  $(L_1 \cup L_2) - Z$ . The vertices of  $Z$ , taken as singleton branch sets, are complete to this model and to one another. They complete a  $K_{r+2}^-$  model.

It remains that  $s = r - 1$ . The union  $X = V(L_1) \cup V(L_2)$  induces a graph containing  $K_{r+1}^-$ . Since an  $(r + 1)$ -connected graph has at least  $r + 2$  vertices,  $G - X$  is nonempty. Let  $C$  be a component of  $G - X$ . If  $C$  missed a vertex of  $X$ , then  $N_G(C)$  would be a cut of order at most  $r$ . Thus  $C$  is adjacent to every vertex of  $X$ . The vertices of  $X$  as singleton branch sets, together with  $C$ , form a  $K_{r+2}^-$  model.  $\square$

The case  $r = 5$  gives the following form used throughout the paper.

**Corollary 2.2.** *Every six-connected  $K_7^-$ -minor-free graph contains at most one  $K_5$  subgraph.*

### 3. DEGREE-SEVEN NEIGHBOURHOODS

We use two standard facts about contraction-critical graphs. A noncomplete  $k$ -contraction-critical graph is seven-connected when  $k \geq 7$  [6]; see also [9, Theorem 1.8]. Also, if  $G$  is  $k$ -contraction-critical and  $v \in V(G)$ , then

$$(3.1) \quad \alpha(G[N(v)]) \leq d(v) - k + 2;$$

This is a standard consequence of contraction-criticality; see Dirac [2] and the formulation in [9, Lemma 1.6(i)].

We shall also use the following specialization of a theorem of Kriesell and Mohr [5, Theorem 7]. Let a graph  $J$  be properly coloured with five colours, and choose one root from each colour class. Let  $D$  be a graph on the five roots with at most six edges. If the ends of every edge  $xy \in E(D)$  lie in the same bichromatic component of  $J$  on their two colours, then  $J$  contains pairwise disjoint connected sets, one containing each root, that are adjacent for every edge of  $D$ .

**Lemma 3.1.** *Let  $G$  be a seven-chromatic graph such that every proper minor is six-colourable and  $K_7^- \not\leq G$ . Let  $v$  have degree seven, put  $H = G[N(v)]$ , and let  $ab$  be a nonedge of  $H$ . If*

$$R = N(v) \setminus \{a, b\},$$

*then  $G - v - \{a, b\}$  contains a  $K_5$  minor rooted at  $R$ .*

*Proof.* Contract the two edges  $va$  and  $vb$  to one vertex. A six-colouring of the resulting proper minor induces a six-colouring  $\varphi$  of  $G - v$  in which  $a$  and  $b$  have one common colour, say 0. Every vertex of  $R$  avoids colour 0. The five vertices of  $R$  use the other five colours distinctly, since a colour absent from  $N(v)$  could be assigned to  $v$ .

Put  $F = \overline{H}$ . Inequality (3.1) gives  $\alpha(H) \leq 2$ , so  $F$  is triangle-free. If  $xy \in E(F[R])$  and  $x, y$  have colours  $i, j$ , then  $x, y$  lie in the same component of the subgraph of  $G - v$  induced by colours  $i, j$ . Otherwise, interchanging  $i$  and  $j$  on the component containing  $x$  would remove colour  $i$  from  $N(v)$ , allowing that colour at  $v$ .

Delete the colour-0 class from  $G - v$ . The resulting graph is properly five-coloured, with  $R$  a transversal of its colour classes. By Mantel's theorem, the triangle-free graph  $F[R]$  has at most six edges. The Kriesell–Mohr result, applied with demand graph  $F[R]$ , gives five disjoint connected sets rooted at  $R$  and adjacent for every nonedge of  $H[R]$ . Every other pair is joined by its edge in  $H[R]$ , so the five sets form the required rooted  $K_5$  model.  $\square$

**Theorem 3.2.** *Under the hypotheses of Lemma 3.1, every degree-seven vertex  $v$  satisfies*

$$(3.2) \quad G[N(v)] \cong K_4 \dot{\cup} K_3 \quad \text{or} \quad K_1 \vee (K_3 \dot{\cup} K_3).$$

*In particular,  $v$  belongs to a  $K_5$  subgraph.*

*Proof.* Put  $H = G[N(v)]$  and  $F = \overline{H}$ . By (3.1),  $F$  is triangle-free. It is nonempty, since otherwise  $G[N(v)]$  would contain  $K_7^-$ .

Every nonisolated vertex of  $F$  has degree at least three. To see this, choose an edge  $ab \in E(F)$  and use the rooted model of Lemma 3.1 on  $R = N(v) \setminus \{a, b\}$ . If  $d_F(a) = 1$ , then  $a$  is adjacent in  $H$  to every root; the rooted model,  $\{a\}$ , and  $\{v\}$  form a  $K_7$  model. If  $d_F(a) = 2$ , with other neighbour  $x$ , then triangle-freeness gives  $bx \in E(H)$ . Absorb  $b$  into the branch set rooted at  $x$ . Together with  $\{a\}$  and  $\{v\}$ , the resulting seven branch sets have at most one missing adjacency, again a contradiction.

Every nontrivial component of  $F$  therefore has minimum degree at least three and hence at least six vertices. Since  $|V(F)| = 7$ , there is one nontrivial component and at most one isolated vertex. If the component has six vertices, Mantel's theorem and its equality case give  $F \cong K_{3,3} \dot{\cup} K_1$ .

Suppose instead that the component has seven vertices. No vertex has degree at least five: its independent neighbourhood would contain a vertex of degree at most two. Thus every degree is three or four. The degree sum is even, so some vertex  $z$  has degree four. Its four neighbours are independent. Each is adjacent to both vertices outside  $N_F[z]$  in order to have degree at least three, and those two outside vertices are nonadjacent by triangle-freeness. Hence  $F \cong K_{3,4}$ . Taking complements gives (3.2).  $\square$

For a minor-minimal non-six-colourable  $K_7^-$ -minor-free graph, Albar's theorem gives  $\chi(G) = 7$ , while Mader's theorem gives seven-connectivity. Corollary 2.2 and Theorem 3.2 then give the sharper conclusion below.

**Corollary 3.3.** *Let  $G$  be as in Theorem 1.2. Then  $G$  contains at most one  $K_5$  subgraph and every degree-seven vertex  $v$  satisfies*

$$G[N(v)] \cong K_4 \dot{\cup} K_3.$$

*Proof.* The only point not already stated is the exclusion of the second graph in (3.2). In that graph,  $v$  belongs to two distinct  $K_5$  subgraphs, which contradicts Corollary 2.2.  $\square$

#### 4. PRIVATE TRIANGLES

The next lemma is the only substantial colouring argument in the paper. Its conclusion is deliberately stated in terms of a  $K_7^-$  model: one of the two nonsingleton branch sets may miss one of the five triangles.

**Lemma 4.1** (Private-triangle allocation). *Let  $G$  be a seven-chromatic graph such that every proper minor of  $G$  is six-colourable. Let  $A$  be a  $K_5$  subgraph of  $G$ . Suppose that, for each  $a \in A$ ,*

$$T_a = N_G(a) \setminus A$$

*is a triangle, that the triangles  $T_a$  are pairwise disjoint, that  $T_a$  is anticomplete to  $A \setminus \{a\}$ , and that  $G - A$  is connected. Then  $K_7^- \preceq G$ .*

The proof fixes one edge-deletion colouring. It first forces two colours onto the three private triangles indexed by  $J$  and records the resulting Kempe reach. A palette obstruction then gives two alternatives; in either one, two disjoint connected sets are allocated so that one meets all five private triangles and the other meets at least four.

*Proof.* Put  $H = G - A$ . Fix  $a_i \in A$  and  $x \in T_{a_i}$ . Let  $G - a_i x$  denote deletion of the edge  $a_i x$ , and take a six-colouring  $\varphi$  of this graph. The vertices  $a_i$  and  $x$  have the same colour, say  $p$ , since otherwise  $\varphi$  would colour  $G$ . The clique  $A$  uses five colours. Denote by  $q$  the colour absent from  $A$ . The colour  $q$  occurs on  $T_{a_i}$ , since otherwise  $a_i$  could be recoloured  $q$ . Thus

$$\varphi(T_{a_i}) = \{p, q, r\}.$$

Let  $a_h$  be the vertex of  $A \setminus \{a_i\}$  coloured  $r$ , put

$$J = A \setminus \{a_i, a_h\}, \quad c_j = \varphi(a_j) \quad (a_j \in J),$$

and write  $p_a, q_a$  for the uniquely  $p$ -coloured and  $q$ -coloured vertices of  $T_a$ , whenever those colours occur there. In particular,  $p_{a_i} = x$ . For colours  $\alpha, \beta$ , let  $H[\alpha, \beta]$  denote the subgraph of  $H$  induced by the vertices of those two colours.

For  $a_j \in J$ , both  $p$  and  $q$  occur on  $T_{a_j}$ . Indeed, if  $\gamma \in \{p, q\}$  were absent from  $T_{a_j}$ , then assigning  $c_j$  to  $a_i$ , assigning  $\gamma$  to  $a_j$ , and retaining the colours on the other three vertices of  $A$  would colour  $G$ . The five assigned colours are distinct,  $c_j$  is absent from  $T_{a_i}$ ,  $\gamma$  is absent from  $T_{a_j}$ , and the restored edge  $a_i x$  has colours  $c_j, p$ .

For  $\gamma \in \{p, q\}$ , let  $C_j^\gamma$  be the component of  $H[\gamma, c_j]$  containing  $\gamma_{a_i}$ . Then

$$(4.1) \quad \gamma_{a_j} \in C_j^\gamma.$$

Suppose otherwise and interchange  $\gamma, c_j$  on  $C_j^\gamma$ . If  $\gamma = p$ , this changes  $x$  from  $p$  to  $c_j$ , so  $a_i$  may be assigned  $p$ . If  $\gamma = q$ , it removes  $q$  from  $T_{a_i}$ , so  $a_i$  may be

assigned  $q$ , while  $x$  remains coloured  $p$ . In either case  $a_j$  retains  $c_j$ : initially  $c_j \notin \varphi(T_{a_j})$ , and the component misses  $T_{a_j}$ . All other retained clique colours lie outside  $\{\gamma, c_j\}$ . Hence the interchange extends to a six-colouring of  $G$ , proving (4.1).

Write

$$\varphi(T_{a_j}) = \{p, q, s_{a_j}\} \quad (a_j \in J).$$

If  $T_{a_h}$  omits some  $\gamma \in \{p, q\}$  and  $s_{a_j} \neq r$  for some  $a_j \in J$ , then  $\gamma$  is available at  $a_h$  and  $r$  is available at  $a_j$ . Assigning

$$a_h \mapsto \gamma, \quad a_j \mapsto r, \quad a_i \mapsto c_j$$

and retaining the other two clique colours again colours  $G$ . Thus one of the following holds:

$$(4.2) \quad \varphi(T_a) \supseteq \{p, q\} \quad \text{for every } a \in A;$$

$$(4.3) \quad \varphi(T_{a_j}) = \{p, q, r\} \quad \text{for every } a_j \in J.$$

Assume first that (4.3) holds. For  $a_j \in J$ ,  $\gamma \in \{p, q\}$ , and  $a_t \in J \setminus \{a_j\}$ , we have

$$(4.4) \quad \gamma_{a_t} \in C_j^\gamma.$$

Otherwise the component  $D$  of  $H[\gamma, c_j]$  containing  $\gamma_{a_t}$  differs from  $C_j^\gamma$ . Since  $C_j^\gamma$  contains the unique  $\gamma$ -coloured vertices of  $T_{a_i}$  and  $T_{a_j}$ , while neither triangle contains  $c_j$ , the component  $D$  misses both triangles. Moreover  $T_{a_t}$  contains no  $c_j$ . Interchange  $\gamma, c_j$  on  $D$ , assign  $\gamma$  to  $a_t$  and  $c_t$  to  $a_i$ , and retain the other clique colours. The interchange removes  $\gamma$  from  $T_{a_t}$ , protects the retained colour  $c_j$  at  $a_j$ , and leaves  $x$  coloured  $p$ . This gives a six-colouring of  $G$ , proving (4.4).

Choose distinct  $a_j, a_\ell \in J$ , and put

$$X = C_j^p, \quad Y = C_\ell^q.$$

The sets  $X$  and  $Y$  are connected and disjoint, since their colour sets  $\{p, c_j\}$  and  $\{q, c_\ell\}$  are disjoint. By (4.1) and (4.4), each meets  $T_{a_i}$  and every  $T_{a_t}$  with  $a_t \in J$ . On each of these four triangles their intersections contain  $p_a$  and  $q_a$ , respectively, so  $X$  and  $Y$  are adjacent. If  $T_{a_h}$  misses  $X \cup Y$ , take in  $H$  a shortest path from  $T_{a_h}$  to  $X \cup Y$ , stopped at its first vertex in that union, and add the path except for its final vertex to the set containing that final vertex. If  $T_{a_h}$  already meets the union, enlarge nothing. We obtain disjoint connected sets  $X', Y'$ , one meeting all five triangles and the other meeting at least four, with  $X'$  adjacent to  $Y'$ . Together with the five singleton branch sets  $\{a\}$ ,  $a \in A$ , they form a  $K_7^-$  model: only the adjacency between  $a_h$  and the set missing  $T_{a_h}$  may be absent.

It remains to assume (4.2) and that (4.3) does not hold. Write

$$\varphi(T_a) = \{p, q, s_a\} \quad (a \in A).$$

Then

$$(4.5) \quad s_{a_i} = r, \quad s_{a_h} \in \{c_j : a_j \in J\}, \quad s_{a_j} \neq c_j \quad (a_j \in J).$$

The first equality is the definition of  $r$ . Every  $a \in A \setminus \{a_i\}$  remains adjacent to all three vertices of  $T_a$  in  $G - a_i x$ , so its colour is absent from  $\varphi(T_a)$ . Applied to  $a_h$  and to the vertices  $a_j \in J$ , this proves the other two assertions in (4.5). Since (4.3) fails, choose  $a_u \in J$  with  $s_{a_u} \neq r$ .

We claim that, for  $a_j \in J$ ,  $\gamma \in \{p, q\}$ , and  $a_t \in A$ ,

$$(4.6) \quad s_{a_t} \neq c_j \implies \gamma_{a_t} \in C_j^\gamma.$$

The cases  $a_t = a_i, a_j$  follow from the definition and (4.1). If  $a_t \in J \setminus \{a_j\}$ , a contrary component  $D$  misses  $T_{a_i}$  and  $T_{a_j}$ , while  $s_{a_t} \neq c_j$  makes  $T_{a_t}$  free of  $c_j$ . Interchanging  $\gamma, c_j$  on  $D$ , assigning  $\gamma$  to  $a_t$  and  $c_t$  to  $a_i$ , and retaining the other clique colours gives the same contradiction as in the proof of (4.4).

It remains to check  $a_t = a_h$ . Let  $D$  be the  $H[\gamma, c_j]$ -component containing  $\gamma_{a_h}$ , and suppose  $D \neq C_j^\gamma$ . Then  $D$  misses  $T_{a_i}$  and  $T_{a_j}$ , and  $s_{a_h} \neq c_j$  makes  $T_{a_h}$  free of  $c_j$ . After interchanging  $\gamma, c_j$  on  $D$ , assign

$$(4.7) \quad a_h \mapsto \gamma, \quad a_i \mapsto c_u, \quad a_u \mapsto r,$$

and retain the original colours on  $J \setminus \{a_u\}$ . The interchange removes  $\gamma$  from  $T_{a_h}$ ; the choice  $s_{a_u} \neq r$  makes  $r$  available at  $a_u$ ; and  $c_u$  is available at  $a_i$ . The five clique colours in (4.7) and on the two retained vertices are distinct. If  $a_u = a_j$ , no clique vertex retains  $c_j$ ; if  $a_u \neq a_j$ , the fact that  $D$  misses  $T_{a_j}$  protects the retained colour  $c_j$ . Every other retained colour lies outside  $\{\gamma, c_j\}$ . Finally,  $D$  misses  $T_{a_i}$ , so  $x$  retains colour  $p$  and the restored edge has colours  $c_u, p$ . This proves (4.6), including the exceptional recolouring.

For  $a_j \in J$ , let

$$m_j = |\{a \in A : s_a = c_j\}|.$$

Since  $s_{a_i} = r$ , we have  $\sum_{a_j \in J} m_j \leq 4$ . Choose  $a_\ell \in J$  with  $m_\ell \leq 1$ , and write  $J = \{a_j, a_k, a_\ell\}$ . Set

$$X = C_j^p \cup C_k^p, \quad Y = C_\ell^q.$$

The set  $X$  is connected because both components contain  $x$ , and  $Y$  is connected. They are disjoint because their colour sets are contained in  $\{p, c_j, c_k\}$  and  $\{q, c_\ell\}$ , respectively. For every  $a \in A$ , the value  $s_a$  cannot equal both  $c_j$  and  $c_k$ , so (4.6) places  $p_a$  in  $X$ . Thus  $X$  meets all five triangles. The same claim places  $q_a$  in  $Y$  unless  $s_a = c_\ell$ , which occurs for at most one  $a$ . Every other triangle supplies the edge  $p_a q_a$  between  $X$  and  $Y$ . Therefore  $X, Y$ , together with the five singleton sets  $\{a\}$ , form a  $K_7^-$  model.  $\square$

**Corollary 4.2.** *Let  $G$  be as in Theorem 1.2. No  $K_5$  subgraph of  $G$  has all five vertices of degree seven.*

*Proof.* Suppose that  $A \cong K_5$  and every vertex of  $A$  has degree seven. Corollary 2.2 makes  $A$  the unique  $K_5$  subgraph. Corollary 3.3 shows that, for each  $a \in A$ ,  $T_a = N(a) \setminus A$  is a triangle anticomplete to  $A \setminus \{a\}$ . The triangles are pairwise disjoint: a vertex in  $T_a \cap T_b$  would be adjacent to  $b$  by its membership in  $T_b$  and nonadjacent to  $b$  by the defining property of



$T_a$ . Finally,  $G - A$  is connected because  $G$  is seven-connected and  $|A| = 5$ . Lemma 4.1 gives the forbidden  $K_7^-$  minor.  $\square$

## 5. NUMERICAL CONSEQUENCES

We first record the density input. Jakobsen proved that an  $n$ -vertex graph with  $n \geq 7$  and at least  $\frac{9}{2}n - 12$  edges contains a  $K_7^-$  minor or is a  $(K_{2,2,2,2}, K_6, 4)$ -cockade [4]; see also [1, Theorem 2].

**Lemma 5.1.** *If  $G$  is as in Theorem 1.2, with  $n = |V(G)|$  and  $m = |E(G)|$ , then*

$$(5.1) \quad 2m \leq 9n - 25.$$

*Proof.* The graph  $G$  is not either base graph of the cockade family, since both  $K_6$  and  $K_{2,2,2,2}$  are six-colourable. A nontrivial cockade has a vertex cut of order four, whereas  $G$  is seven-connected. Thus  $G$  is not an exceptional cockade. Since it has no  $K_7^-$  minor, Jakobsen's theorem gives  $m < \frac{9}{2}n - 12$ . Integrality yields (5.1).  $\square$

**Proposition 5.2.** *Let  $G$  be as in Theorem 1.2. Then*

$$(5.2) \quad n_7 \leq 4, \quad |E(G)| \geq 4|V(G)| - 2,$$

and

$$(5.3) \quad b \geq 20 - n_7 + \tau.$$

*Proof.* Every degree-seven vertex lies in the unique possible  $K_5$  by Corollary 3.3, and Corollary 4.2 permits at most four such vertices. Every other vertex has degree at least eight, so

$$2|E(G)| \geq 7n_7 + 8(|V(G)| - n_7) = 8|V(G)| - n_7,$$

which proves (5.2).

Every degree-seven vertex and every degree-eight vertex not counted by  $b$  lies in the unique possible  $K_5$ . Hence

$$(5.4) \quad n_7 + (n_8 - b) \leq 5.$$

The exact degree identity and Lemma 5.1 give

$$(5.5) \quad 25 \leq 9|V(G)| - 2|E(G)| = 2n_7 + n_8 - \tau.$$

Combining (5.4) and (5.5) gives (5.3).  $\square$

The bound (5.3) gives only  $16 + \tau$  when  $n_7 = 4$ . In that case, contracting the unique clique improves the upper density bound enough to remove this loss.

**Proposition 5.3.** *If  $G$  is as in Theorem 1.2 and  $n_7 = 4$ , then*

$$(5.6) \quad 2|E(G)| \leq 9|V(G)| - 41, \quad |V(G)| \geq 37, \quad n_8 \geq 33 + \tau, \quad b \geq 32 + \tau.$$

*Proof.* All four degree-seven vertices lie in the unique  $K_5$  subgraph. Write it as

$$A = \{a_1, a_2, a_3, a_4, z\}.$$

For  $1 \leq i \leq 4$ , Corollary 3.3 gives

$$(5.7) \quad N(a_i) = (A \setminus \{a_i\}) \cup T_i,$$

where  $T_i$  is a triangle. These four triangles are pairwise disjoint, and  $z$  is anticomplete to all of them.

Put  $H = G - A$ . We claim that  $H$  is four-connected. Suppose that  $S \subseteq V(H)$ , with  $|S| \leq 3$ , leaves components  $C_1, \dots, C_q$ , where  $q \geq 2$ . Seven-connectivity gives

$$(5.8) \quad |N_A(C_j)| \geq 7 - |S| \quad (1 \leq j \leq q).$$

Each  $a_i$  has neighbours in at most one component, since  $T_i - S$  is connected whenever it is nonempty. The vertex  $z$  has neighbours in at most all  $q$  components. Counting incidences between  $A$  and the components in (5.8) gives

$$q(7 - |S|) \leq 4 + q,$$

which is impossible for  $q \geq 2$  and  $|S| \leq 3$ . This proves the claim.

Contract  $A$  to one vertex, and call the resulting graph  $J$ . No vertex outside  $A$  has two neighbours in  $A$ : any outside neighbour of  $a_i$  lies in  $T_i$  and is anticomplete to  $A \setminus \{a_i\}$ . Thus the contraction creates no parallel edges, and

$$(5.9) \quad |V(J)| = |V(G)| - 4, \quad |E(J)| = |E(G)| - 10.$$

The graph  $J$  is five-connected. A cut of order at most four avoiding the contracted vertex would also separate  $G$ ; a cut containing it would leave  $H$  minus at most three vertices disconnected.

The four disjoint triangles imply  $|V(J)| \geq 13$ . Hence  $J$  is neither  $K_6$  nor  $K_{2,2,2,2}$ . It is not a nontrivial  $(K_{2,2,2,2}, K_6, 4)$ -cockade, since such a graph has a cut of order four. It is also  $K_7^-$ -minor-free, being a minor of  $G$ . Jakobsen's theorem and integrality give

$$2|E(J)| \leq 9|V(J)| - 25.$$

Substituting (5.9) proves the first inequality in (5.6). Proposition 5.2 gives  $2|E(G)| \geq 8|V(G)| - 4$ , so  $|V(G)| \geq 37$ .

Finally, (5.5) with its improved left side gives

$$41 \leq 9|V(G)| - 2|E(G)| = 8 + n_8 - \tau.$$

Thus  $n_8 \geq 33 + \tau$ . Inequality (5.4) and  $n_7 = 4$  show that at most one degree-eight vertex is not counted by  $b$ , so  $b \geq 32 + \tau$ .  $\square$

*Proof of Theorem 1.2.* Albar's theorem gives  $\chi(G) = 7$ , and every proper minor is six-colourable by the choice of  $G$ . The graph is noncomplete because it has no  $K_7^-$  minor, so Mader's theorem gives seven-connectivity.

Corollaries 2.2 and 3.3 give the clique and neighbourhood conclusions, and Corollary 4.2 excludes five degree-seven vertices in the clique.

If  $n_7 \leq 3$ , the degree sum is an even integer at least  $8|V(G)| - 3$ , so  $|E(G)| \geq 4|V(G)| - 1$ ; (5.3) gives  $b \geq 17 + \tau$ . If  $n_7 = 4$ , Proposition 5.3 gives the second alternative. The three displayed global conclusions follow.  $\square$

Theorem 1.2 gives the following reduction of the Norin–Totschnig conjecture to an extremal minor problem.

**Corollary 5.4** (Extremal reduction). *Suppose that every seven-connected graph  $H$  satisfying*

$$|E(H)| \geq 4|V(H)| - 2$$

*contains a  $K_7^-$  minor. Then every  $K_7^-$ -minor-free graph is six-colourable.*

*Proof.* Otherwise, choose a minor-minimal non-six-colourable  $K_7^-$ -minor-free graph  $G$ . By Theorem 1.2, the graph  $G$  is seven-connected and  $|E(G)| \geq 4|V(G)| - 2$ . The assumed extremal statement therefore gives  $K_7^- \preceq G$ , a contradiction.  $\square$

The extremal statement in Corollary 5.4 remains open. Thus the present paper does not prove the Norin–Totschnig conjecture, and it does not settle Hadwiger’s conjecture for chromatic number seven.

#### STATEMENT ON AI USE AND INDEPENDENCE

This research was conducted independently by Cavit Erginsoy using GPT-5.6 Sol, a model developed by OpenAI. GPT-5.6 Sol contributed substantially to theorem discovery, proof development, verification, and manuscript preparation. Erginsoy directed and curated the research programme. Erginsoy has no affiliation with OpenAI; OpenAI did not sponsor, supervise, endorse, or verify this work.

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